Exercise 10.3: Max Version of Hilbert's Inequality

A square-root weight, Cauchy–Schwarz, and a sharp constant

A step-by-step tutorial for a high-school reader

What you will learn

By the end of this tutorial, you will be able to:

- read the double sum as a weighted matrix expression;
- see why the line $m = n$ divides the problem into two simple regions;
- prove two useful estimates involving square roots;
- use a carefully chosen weight $w_n = 1/\sqrt{n}$ inside Cauchy–Schwarz;
- prove the constant 4 is the smallest possible universal constant.

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1 The exercise, stated carefully

The problem

Let $a = (a_1, a_2, \dots)$ and $b = (b_1, b_2, \dots)$ be real sequences whose squares have finite sums:

$$\sum_{m=1}^{\infty} a_m^2 < \infty, \qquad \sum_{n=1}^{\infty} b_n^2 < \infty.$$

Show that

$$\boxed{\left| \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} \frac{a_m b_n}{\max\{m,n\}} \right| < 4 \left(\sum_{m=1}^{\infty} a_m^2 \right)^{1/2} \left(\sum_{n=1}^{\infty} b_n^2 \right)^{1/2}} \tag{1}$$

when both sequences are nonzero. Then prove that no constant smaller than 4 can work for every pair of sequences.

The displayed exercise does not put absolute-value signs around the double sum. We will prove the stronger estimate (1). Since every real number S satisfies $S \le |S|$, the requested inequality follows immediately.

A small point about the strict inequality

If one sequence is identically zero, then both sides are zero, so the literal statement $0 < 0$ is false. The fully precise version is

$$|S(a,b)| \le 4 \|a\|_2 \|b\|_2$$

for all square-summable sequences, with strict inequality when both a and b are nonzero. This is the version proved below.

1.1 The infinite-vector length

For a finite vector, Pythagoras gives the length

$$\sqrt{a_1^2 + a_2^2 + \cdots + a_N^2}.$$

For an infinite sequence, we use the same idea:

$$\|a\|_2 := \left(\sum_{m=1}^{\infty} a_m^2 \right)^{1/2}.$$

A sequence with $\|a\|_2 < \infty$ is called *square-summable*, or an ℓ^2 sequence.

1.2 The kernel matrix

Set

$$K_{mn} := \frac{1}{\max\{m,n\}}.$$

Then the double sum is

$$S(a,b) = \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} K_{mn} a_m b_n.$$

The beginning of the matrix (K_{mn}) is

$$\begin{pmatrix} 1 & \frac{1}{2} & \frac{1}{3} & \frac{1}{4} & \frac{1}{5} & \cdots \\ \frac{1}{2} & \frac{1}{2} & \frac{1}{3} & \frac{1}{4} & \frac{1}{5} & \cdots \\ \frac{1}{3} & \frac{1}{3} & \frac{1}{3} & \frac{1}{4} & \frac{1}{5} & \cdots \\ \frac{1}{4} & \frac{1}{4} & \frac{1}{4} & \frac{1}{4} & \frac{1}{5} & \cdots \\ \frac{1}{5} & \frac{1}{5} & \frac{1}{5} & \frac{1}{5} & \frac{1}{5} & \cdots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \ddots \end{pmatrix}.$$

Notice that the matrix is symmetric: $K_{mn} = K_{nm}$.

2 The geometry hidden inside $\max\{m, n\}$

The line $m = n$ divides the grid of pairs (m, n) into two regions:

$$K_{mn} = \frac{1}{\max\{m,n\}} = \begin{cases} \dfrac{1}{m}, & m \geq n, \\ \dfrac{1}{n}, & n > m. \end{cases} \tag{2}$$

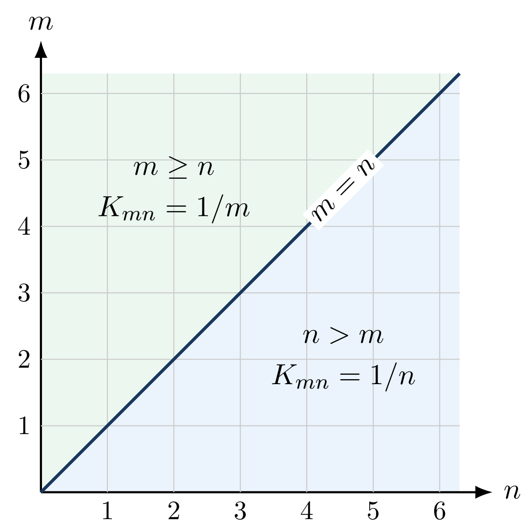


Figure 1. The maximum becomes simple on each side of the diagonal.

This split is the first major idea. For a fixed row number m, the terms with $n \leq m$ behave differently from the terms with $n > m$.

3 Three tools we need

3.1 Cauchy–Schwarz

For finite lists of real numbers,

$$\left|\sum_j x_j y_j\right| \leq \left(\sum_j x_j^2\right)^{1/2} \left(\sum_j y_j^2\right)^{1/2}. \tag{3}$$

The same formula works for convergent infinite sums. It also works when the index j is replaced by a pair (m, n): simply imagine writing all grid points in one long list.

Why ordinary Cauchy–Schwarz is not enough

A first attempt might be to separate $a_m b_n$ from the kernel K_{mn}. That leads to the sum $\sum_{m,n} K_{mn}^2$, which diverges. The successful proof puts a square-root weight into Cauchy–Schwarz so that each row becomes manageable.

3.2 A telescoping sum

A sum is called *telescoping* when most terms cancel. For example,

$$\begin{aligned}(\sqrt{1} - \sqrt{0}) + (\sqrt{2} - \sqrt{1}) + \cdots + (\sqrt{m} - \sqrt{m-1})\\ = \sqrt{m}.\end{aligned}$$

We will compare our sums with expressions of this form.

3.3 Two square-root estimates

Estimate A: the first part of a row

For every integer $m \geq 1$,

$$\boxed{\sum_{n=1}^{m} \frac{1}{\sqrt{n}} < 2\sqrt{m}.} \tag{4}$$

Indeed,

$$2(\sqrt{n} - \sqrt{n-1}) = \frac{2}{\sqrt{n} + \sqrt{n-1}} > \frac{1}{\sqrt{n}},$$

because $\sqrt{n} + \sqrt{n-1} < 2\sqrt{n}$. Summing from $n = 1$ to $n = m$ makes the right-hand comparison telescope:

$$\sum_{n=1}^{m} \frac{1}{\sqrt{n}} < 2\sum_{n=1}^{m} (\sqrt{n} - \sqrt{n-1}) = 2\sqrt{m}.$$

Estimate B: the tail of a row

For every integer $m \geq 1$,

$$\boxed{\sum_{n=m+1}^{\infty} \frac{1}{n^{3/2}} < \frac{2}{\sqrt{m}}.} \tag{5}$$

For $n \geq 2$,

$$\begin{aligned}2\left(\frac{1}{\sqrt{n-1}} - \frac{1}{\sqrt{n}}\right) &= \frac{2}{\sqrt{n-1}\sqrt{n}(\sqrt{n} + \sqrt{n-1})}\\ &> \frac{1}{n^{3/2}}.\end{aligned}$$

The last inequality follows because each $\sqrt{n-1}$ is smaller than $\sqrt{n}$, so the denominator on the first line is smaller than $2n^{3/2}$. Therefore, for any $M > m$,

$$\begin{aligned}\sum_{n=m+1}^{M} \frac{1}{n^{3/2}} &< 2 \sum_{n=m+1}^{M} \left(\frac{1}{\sqrt{n-1}} - \frac{1}{\sqrt{n}}\right) \\ &= 2\left(\frac{1}{\sqrt{m}} - \frac{1}{\sqrt{M}}\right) < \frac{2}{\sqrt{m}}.\end{aligned}$$

Letting M grow gives (5).

4 The decisive weight and the number 4

Choose the positive weight

$$\boxed{w_n = \frac{1}{\sqrt{n}}.} \tag{6}$$

For a fixed row m, define its weighted row sum by

$$R_m := \sum_{n=1}^{\infty} K_{mn} w_n.$$

Using the split (2),

$$\begin{aligned}R_m &= \sum_{n=1}^{m} \frac{1}{m}\frac{1}{\sqrt{n}} + \sum_{n=m+1}^{\infty} \frac{1}{n}\frac{1}{\sqrt{n}} \\ &= \frac{1}{m}\sum_{n=1}^{m} \frac{1}{\sqrt{n}} + \sum_{n=m+1}^{\infty} \frac{1}{n^{3/2}}.\end{aligned} \tag{7}$$

Apply (4) and (5):

$$\begin{aligned}R_m &< \frac{1}{m}(2\sqrt{m}) + \frac{2}{\sqrt{m}} \\ &= \frac{4}{\sqrt{m}} = 4w_m.\end{aligned}$$

Thus we have proved the central estimate

$$\boxed{R_m < 4w_m \qquad (m = 1, 2, \dots).} \tag{8}$$

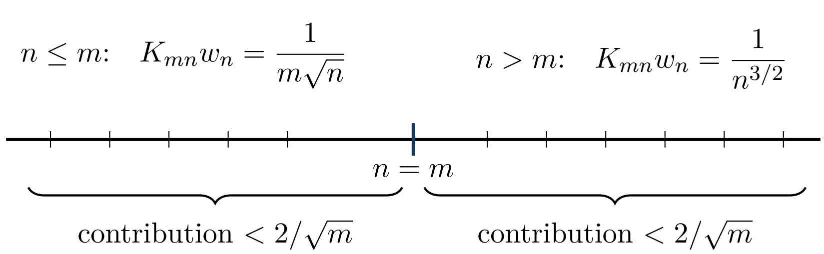


Figure 2. The constant 4 appears as $2 + 2$, one contribution from each side of $n = m$.

Where did the square-root weight come from?

The weight $n^{-1/2}$ is the natural balance point. Roughly speaking, the left part of (7) and the right part both have size $m^{-1/2}$. Neither side dominates the other. Later, the sharpness example uses exactly the same shape $1/\sqrt{n}$, which is strong evidence that this is the correct choice.

5 Proof of the inequality

We now combine the weighted row estimate with Cauchy–Schwarz.

Step 1: remove signs

Because $K_{mn} > 0$,

$$\begin{aligned} |S(a,b)| &= \left| \sum_{m,n\geq 1} K_{mn} a_m b_n \right| \\ &\leq \sum_{m,n\geq 1} K_{mn} |a_m||b_n|. \end{aligned}$$

Call the last sum T. Proving $T < 4\left\|a\right\|_2 \left\|b\right\|_2$ is enough.

Step 2: insert a factor equal to 1

For every pair (m,n),

$$\sqrt{\frac{w_n}{w_m}}\sqrt{\frac{w_m}{w_n}} = 1.$$

Therefore,

$$K_{mn}|a_m||b_n| = \left(\sqrt{K_{mn}}|a_m|\sqrt{\frac{w_n}{w_m}}\right)\left(\sqrt{K_{mn}}|b_n|\sqrt{\frac{w_m}{w_n}}\right).$$

This strange-looking insertion is designed so that the row sum R_m will appear after squaring.

Step 3: apply Cauchy–Schwarz over all pairs (m,n)

Equation (3) gives

$$T \leq \left(\sum_{m,n\geq 1} K_{mn} a_m^2 \frac{w_n}{w_m}\right)^{1/2} \left(\sum_{m,n\geq 1} K_{mn} b_n^2 \frac{w_m}{w_n}\right)^{1/2}. \tag{9}$$

We simplify the first factor by grouping terms with the same m:

$$\begin{aligned} \sum_{m,n\geq 1} K_{mn} a_m^2 \frac{w_n}{w_m} &= \sum_{m=1}^{\infty} \frac{a_m^2}{w_m} \sum_{n=1}^{\infty} K_{mn} w_n \\ &= \sum_{m=1}^{\infty} \frac{a_m^2}{w_m} R_m \\ &< 4 \sum_{m=1}^{\infty} a_m^2. \end{aligned} \tag{10}$$

The last line uses $R_m < 4w_m$. Because the kernel is symmetric, exactly the same argument gives

$$\sum_{m,n\geq 1} K_{mn} b_n^2 \frac{w_m}{w_n} < 4 \sum_{n=1}^{\infty} b_n^2. \tag{11}$$

Substitute (10) and (11) into (9):

$$\begin{aligned} T &< \left(4 \sum_{m=1}^{\infty} a_m^2\right)^{1/2} \left(4 \sum_{n=1}^{\infty} b_n^2\right)^{1/2} \\ &= 4 \left\|a\right\|_2 \left\|b\right\|_2 . \end{aligned}$$

Hence

$$\boxed{\left|\sum_{m=1}^{\infty} \sum_{n=1}^{\infty} \frac{a_m b_n}{\max\{m, n\}}\right| < 4 \left\|a\right\|_2 \left\|b\right\|_2 .}$$

5.1 Why the infinite double sum is legitimate

The two squared sums on the right side of (9) are finite by (10) and (11). Cauchy–Schwarz therefore shows that

$$\sum_{m,n\geq 1} K_{mn} |a_m||b_n| < \infty.$$

So the original double series converges absolutely. One may also apply the argument first to a finite $N \times N$ square and then let $N \to \infty$.

6 What does it mean for 4 to be best possible?

Suppose somebody claims that a smaller number, such as 3.99, works for every pair of sequences. To disprove the claim, we do not need equality at 4. We only need examples whose ratio

$$\frac{\displaystyle\sum_{m,n\geq 1} \frac{a_m b_n}{\max\{m, n\}}}{\left\|a\right\|_2 \left\|b\right\|_2}$$

gets as close to 4 as we wish.

Sharp or best constant

The constant 4 is *sharp* if the inequality is true with 4, but false with every constant $C < 4$.

The weight used in the proof suggests the right test sequence. The infinite sequence $1/\sqrt{n}$ itself is not square-summable, so we cut it off after N terms.

7 The near-equality sequences

For each positive integer N, define

$$c_n^{(N)} = \begin{cases} \dfrac{1}{\sqrt{n}}, & 1 \le n \le N, \\ 0, & n > N. \end{cases} \tag{12}$$

Use the same sequence on both sides: $a = b = c^{(N)}$.

7.1 Their lengths

Let

$$H_N := 1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{N}$$

be the N-th harmonic number. Then

$$\left\| c^{(N)} \right\|_2^2 = \sum_{n=1}^{N} \left(\frac{1}{\sqrt{n}} \right)^2 = H_N.$$

Therefore,

$$\|a\|_2 \, \|b\|_2 = H_N. \tag{13}$$

7.2 The double sum for these sequences

Write

$$Q_N := \sum_{m=1}^{N} \sum_{n=1}^{N} \frac{1}{\sqrt{m}\sqrt{n}\max\{m,n\}}.$$

The summand is symmetric in m, n. We may double the lower triangle $n \le m$, but then the diagonal has been counted twice, so we subtract it once:

$$\begin{aligned} Q_N &= 2 \sum_{m=1}^{N} \sum_{n=1}^{m} \frac{1}{m^{3/2}\sqrt{n}} - \sum_{m=1}^{N} \frac{1}{m^2} \\ &= 2 \sum_{m=1}^{N} \frac{1}{m^{3/2}} \left(\sum_{n=1}^{m} \frac{1}{\sqrt{n}} \right) - \sum_{m=1}^{N} \frac{1}{m^2}. \end{aligned} \tag{14}$$

Why subtract the diagonal?

Doubling one triangular half counts every off-diagonal pair exactly once in each direction, which is correct. But it counts (m, m) twice, while the original square contains it only once. The term on the diagonal is $1/m^2$.

8 Show that the ratio approaches 4

We need a lower estimate for the inner sum in (14). For every $n \ge 1$,

$$\frac{1}{\sqrt{n}} > 2(\sqrt{n+1} - \sqrt{n}),$$

because

$$2(\sqrt{n+1}-\sqrt{n}) = \frac{2}{\sqrt{n+1}+\sqrt{n}} < \frac{1}{\sqrt{n}}.$$

After summing,

$$\begin{aligned} \sum_{n=1}^{m} \frac{1}{\sqrt{n}} &> 2(\sqrt{m+1}-1) \\ &\geq 2\sqrt{m}-2. \end{aligned} \tag{15}$$

Insert (15) into (14):

$$\begin{aligned} Q_N &> 2\sum_{m=1}^{N} \frac{1}{m^{3/2}}(2\sqrt{m}-2) - \sum_{m=1}^{N} \frac{1}{m^2} \\ &= 4H_N - 4\sum_{m=1}^{N} \frac{1}{m^{3/2}} - \sum_{m=1}^{N} \frac{1}{m^2}. \end{aligned} \tag{16}$$

The two error sums stay bounded as N grows. In fact,

$$\sum_{m=1}^{\infty} \frac{1}{m^{3/2}} < 3 \quad \text{and} \quad \sum_{m=1}^{\infty} \frac{1}{m^2} < 2. \tag{17}$$

The first bound follows from Estimate B with $m = 1$, together with the first term 1. For the second, when $m \geq 2$,

$$\frac{1}{m^2} < \frac{1}{m(m-1)} = \frac{1}{m-1} - \frac{1}{m},$$

and the resulting sum telescopes.

Equations (16) and (17) give the simple lower bound

$$\boxed{Q_N > 4H_N - 14.} \tag{18}$$

8.1 Why H_N grows without bound

Group the harmonic series into blocks whose lengths double:

$$1 + \frac{1}{2} + \left(\frac{1}{3} + \frac{1}{4}\right) + \left(\frac{1}{5} + \cdots + \frac{1}{8}\right) + \cdots.$$

In the block ending at 2^j, there are 2^{j-1} terms, and every term is at least $1/2^j$. Thus every such block contributes at least $1/2$. More formally,

$$\begin{aligned} H_{2^r} &= 1 + \sum_{j=1}^{r} \sum_{n=2^{j-1}+1}^{2^j} \frac{1}{n} \\ &\geq 1 + \sum_{j=1}^{r} 2^{j-1}\frac{1}{2^j} = 1 + \frac{r}{2}. \end{aligned} \tag{19}$$

Therefore $H_N \to \infty$.

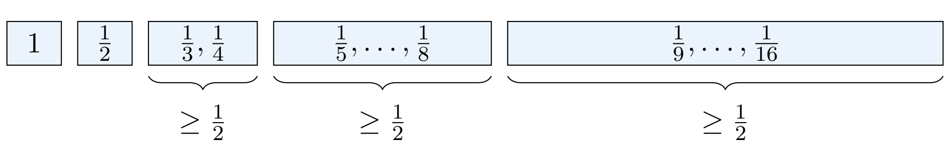


Figure 3. Infinitely many blocks each contribute at least $1/2$, so the harmonic sums grow forever.

8.2 Finish the sharpness proof

For our test sequences, the denominator is H_N by (13). The proved inequality gives $Q_N/H_N < 4$, while (18) gives

$$\frac{Q_N}{H_N} > 4 - \frac{14}{H_N}.$$

Since $H_N \to \infty$,

$$\boxed{\lim_{N\to\infty} \frac{Q_N}{H_N} = 4.} \tag{20}$$

Now let $C < 4$. Choose N so large that

$$\frac{14}{H_N} < 4 - C.$$

Then

$$\frac{Q_N}{H_N} > C,$$

which means

$$Q_N > C \left\| c^{(N)} \right\|_2^2.$$

So an inequality with the smaller constant C fails. Therefore 4 cannot be replaced by any smaller universal constant.

The important distinction

The constant 4 is approached but not attained by nonzero square-summable sequences. The borderline shape $1/\sqrt{n}$ would produce equality-like behavior, but

$$\sum_{n=1}^{\infty} \left(\frac{1}{\sqrt{n}}\right)^2 = \sum_{n=1}^{\infty} \frac{1}{n} = \infty,$$

so it is not an ℓ^2 sequence. Its finite truncations get closer and closer to the sharp constant.

8.3 A numerical look

The approach to 4 is slow because H_N grows only logarithmically. Here are some values of Q_N/H_N:

N	10	100	1,000	10,000	1,000,000
Q_N/H_N	1.978	2.623	2.993	3.217	3.464

The table is not the proof; equation (20) is the proof. The table simply illustrates the slow convergence.

9 A compact homework-ready proof

Main inequality

Let

$$K_{mn} = \frac{1}{\max\{m,n\}}, \qquad w_n = n^{-1/2}.$$

For every $m \geq 1$,

$$\sum_{n=1}^{\infty} K_{mn} w_n = \frac{1}{m} \sum_{n=1}^{m} n^{-1/2} + \sum_{n=m+1}^{\infty} n^{-3/2}$$
$$< \frac{1}{m}(2\sqrt{m}) + \frac{2}{\sqrt{m}} = 4w_m.$$

Therefore, by Cauchy–Schwarz,

$$\sum_{m,n\geq 1} K_{mn}|a_m||b_n| \leq \left(\sum_{m,n\geq 1} K_{mn} a_m^2 \frac{w_n}{w_m} \right)^{1/2} \left(\sum_{m,n\geq 1} K_{mn} b_n^2 \frac{w_m}{w_n} \right)^{1/2}$$
$$< \left(4 \sum_m a_m^2 \right)^{1/2} \left(4 \sum_n b_n^2 \right)^{1/2}$$
$$= 4 \|a\|_2 \|b\|_2 .$$

This proves the stronger absolute-value inequality.

Sharpness

Take $a_n = b_n = n^{-1/2}$ for $n \leq N$, and $a_n = b_n = 0$ for $n > N$. Then $\|a\|_2 \|b\|_2 = H_N$, and

$$Q_N = 2 \sum_{m=1}^{N} m^{-3/2} \sum_{n=1}^{m} n^{-1/2} - \sum_{m=1}^{N} m^{-2}$$
$$> 4H_N - 14.$$

Since $H_N \to \infty$,

$$\frac{Q_N}{\|a\|_2 \|b\|_2} = \frac{Q_N}{H_N} \longrightarrow 4.$$

Thus every constant $C < 4$ fails for all sufficiently large N, so 4 is sharp.

10 Common questions and mistakes

Mistake 1: forgetting the two regions

The formula is not $1/m$ everywhere and not $1/n$ everywhere. Always split at $n = m$:

$$\frac{1}{\max\{m,n\}} = \begin{cases} 1/m, & n \leq m, \\ 1/n, & n > m. \end{cases}$$

Mistake 2: applying Cauchy–Schwarz without a weight

The proof depends on inserting

$$\sqrt{w_n/w_m}\sqrt{w_m/w_n} = 1.$$

Without this balancing factor, the kernel does not have a finite square sum over the whole grid.

Mistake 3: saying "sharp" means equality occurs

A sharp constant need not be attained. Here, nonzero ℓ^2 sequences satisfy a strict inequality, but truncations of $1/\sqrt{n}$ make the ratio tend to 4.

Mistake 4: forgetting to subtract the diagonal

When a symmetric square is written as twice one triangle, the diagonal is counted twice. In the sharpness calculation, subtract $\sum_{m=1}^{N} 1/m^2$ once.

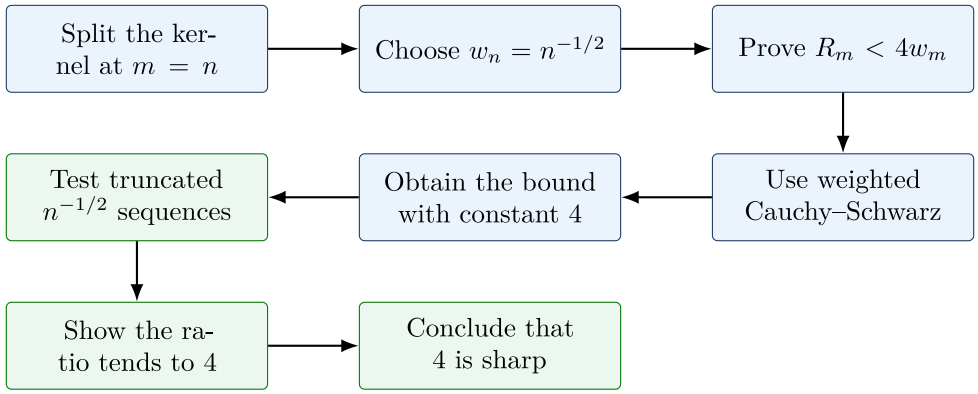
11 Quick self-check

1. Write the 3×3 matrix $(1/\max\{m, n\})$.
2. Why does $\sum_{n=1}^{m} n^{-1/2} < 2\sqrt{m}$ telescope?
3. For a fixed m, what are the two pieces of R_m?
4. Why is the test sequence $1/\sqrt{n}$ cut off after N terms?
5. Explain in one sentence why (20) rules out every constant smaller than 4.

Answers

1. $\begin{pmatrix} 1 & 1/2 & 1/3 \\ 1/2 & 1/2 & 1/3 \\ 1/3 & 1/3 & 1/3 \end{pmatrix}$.
2. Compare $1/\sqrt{n}$ with $2(\sqrt{n} - \sqrt{n-1})$; adjacent square-root terms cancel when summed.
3. $m^{-1} \sum_{n \le m} n^{-1/2}$ and $\sum_{n>m} n^{-3/2}$.
4. The full sequence is not square-summable because its squared terms form the divergent harmonic series.
5. Ratios can be made arbitrarily close to 4, so any fixed $C < 4$ is eventually exceeded.

12 Final map of the proof



The whole argument in one sentence

The square-root weight makes each row of the max-kernel cost less than 4, weighted Cauchy–Schwarz turns that row estimate into the desired inequality, and truncated copies of the same square-root weight show that the number 4 cannot be improved.